

# Shift Manager App – PRD

2026-09-20 · @Someone

## Overview and goals

The shift manager builds the duty detail (who does which role) by hand for each four-day working block. This app lets them set attendance, press Generate, and get a fair, rule-compliant roster that prints in colour for the daily meeting.

### Goals:

- Cut the time spent building each four-day duty detail.
- Never assign someone to a role they are not qualified for.
- Rotate roles fairly, so people do not get the same job or the same hard jobs repeatedly.
- Keep a log of every roster and produce a colour printout for daily meetings.

## Users and scope

Primary user: the shift manager, who builds the roster at a desk on the last night of each block. Everyone else only reads the printed rota.

### In scope:

- 23 people on a 4-on/4-off pattern, with day and night shifts (two days then two nights within the four working days).
- 16 unique roles. Day fills all 16, night fills 14.
- Attendance status, qualification checklist per person, fixed-role people, roster generation, log, colour print.

### Out of scope for now:

- Automatic reading of leave from RDMS (leave is checked by eye and entered by hand).
- Shift swaps, overtime and pay.
- Rosters more than one four-day block ahead.

## Functional requirements

| ID | Requirement                                    | Priority |
|----|------------------------------------------------|----------|
| F1 | Add, edit and deactivate people (currently 23) | Must     |

| ID  | Requirement                                                                            | Priority |
|-----|----------------------------------------------------------------------------------------|----------|
| F2  | Per person, a checklist of the 16 roles they are qualified for                         | Must     |
| F3  | Per person, an optional “only do this role” setting, chosen from their qualified roles | Must     |
| F4  | Set each person’s status for each day: Present, Annual leave, Sick leave, Duty away    | Must     |
| F5  | Generate button fills roles for all Present people for the selected four-day block     | Must     |
| F6  | Generation only assigns qualified people to a role (hard block)                        | Must     |
| F7  | Roles are shuffled and rotated so the same job and pairings do not repeat              | Must     |
| F8  | Hard roles are not given on back-to-back nights, especially the last two               | Must     |
| F9  | Manager can manually change any generated assignment                                   | Must     |
| F10 | Every saved roster is logged and can be looked up later                                | Must     |
| F11 | Print the four-day rota in colour, with paired roles sharing a colour                  | Must     |
| F12 | Manage the role list: name, shift use, required group, hard or easy flag               | Should   |
| F13 | Warn when a role cannot be filled because nobody qualified is Present                  | Should   |

## Business rules

1. **Pattern:** 4 days on, then 4 days off. The four working days are two days then two nights.
2. **Roles:** 16 unique roles. Day fills all 16. Night fills 14, as 2 roles are not used at night.
3. **Skill-restricted roles:** Gulf Sulu 101, Gulf Sulu 103, Member in Charge (MIC) and Jailer can only be filled by people ticked as qualified. This is a hard block.
4. **General roles:** anyone can do them, but each person’s qualification checklist still applies, so it can be limited.

5. **Fixed role:** if a person has "only do this role" and is Present, they always get that role.
6. **Attendance:** only people with status Present are allocated. Annual leave, sick leave and duty away are shown but not allocated.
7. **Rotation:** the role order is shuffled each time, and a person should not get the same role as their previous working day where an alternative exists.
8. **Hard roles:** roles flagged hard (busy until about 6 to 7:15 a.m.) are not given on consecutive nights. Easy roles (finish about 4 a.m.) can follow.
9. **Planning horizon:** rosters are built for one four-day block at a time.

## Non-functional requirements, assumptions and open questions

### Non-functional:

- Data lives in Excel tables, so the manager can edit it directly and Power Apps can connect to it later.
- Generating a four-day roster takes seconds.
- The colour printout fits on one page per shift.

### Assumptions:

- One person edits the roster at a time.
- Excel is the first front end, with the tables designed so Power Apps could replace it later.
- The whole 23-person group is rostered together, with the on-duty crew for each block chosen by the manager through the status dropdown.

### Open questions:

- The full list of 16 roles, their groups, and which are hard or easy.
- Which 2 roles are not filled at night.
- How the 23 people split into teams on the 4-on/4-off cycle.
- Whether skills expire.
- Whether the log must record who edited what and when.
- Whether other people need to view the rota on screen rather than on paper.